

Subject:

Multiplexing :-

→ Module → Multiplexer

Frequency
Wavelength } Analog

Time Division } Digital

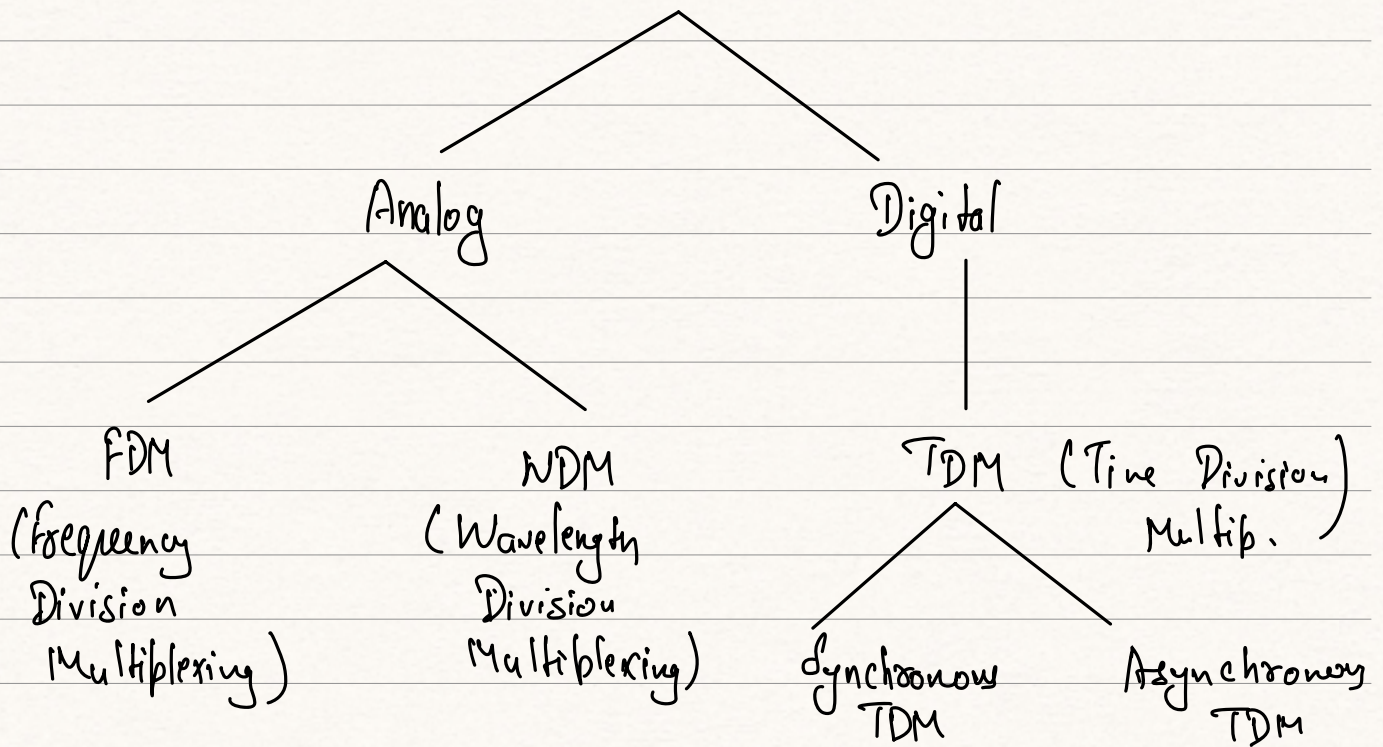
↓ ↓
Synchronous Asynchronous.

* Multiplexer

→ Accept multiple signal and convert in single signal
→ Use two component Mux, Demux

↳ multiply single signal to multiple signal.

Subject: / /

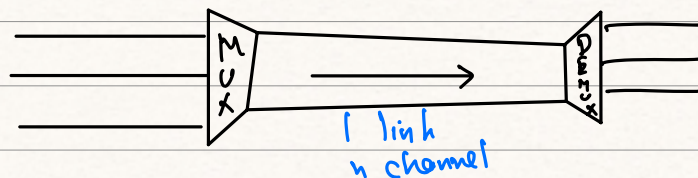


What is Multiplexing?

Process of combining multiple signal into one signal over a shared medium.

→ It is a technique used to combine and send the multiple data stream over a single medium

→ Hardware used for multiplexing is MUX



→ The transmission medium is used to send the signal from sender to receiver. The medium can only have one signal at a time.

Subject: $\rightarrow$ Multiple signal to share one medium, / /
medium must be divided in such a way that each signal is given some portion of the available bandwidth.



$\rightarrow$ uses various frequencies ^{freq} to combine stream of data.

$\rightarrow$ for sending them on a communication medium as a single signal

Main Aim: of FDM is to subdivide the available bandwidth into different frequencies, channels & allocate them to diff. devices.

Advantage $\rightarrow$ simple and easy modulation
 $\rightarrow$ Doesn't require any synchronization b/w sender and receiver.
 $\rightarrow$ Suffer from the problem of cross talk
 $\rightarrow$ large number of modulators are required

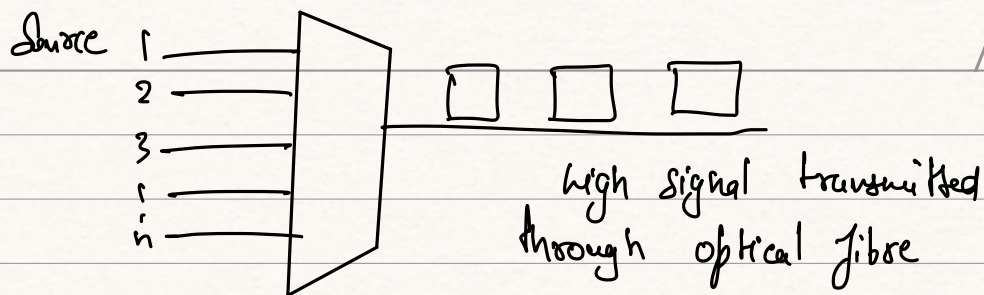
WDM:

is same as FDM except that optical signal are transmitted through the fiber optic cable

$\rightarrow$ increase the capacity of single fiber

$\rightarrow$ Multiplexing and Demultiplexing can be achieved using a prism.

Subject:

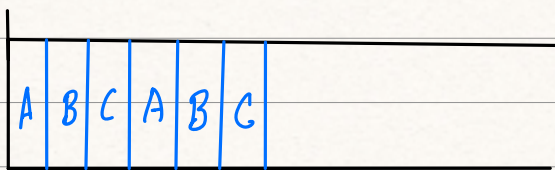


3. TDM:- In FDM - all signal operate at the same time with diff. frequency

but in case of TDM technique all signal operate at same freq. with different time.

→ Total time available in ^{the} channel is distributed among different users.

→ each user is allocated with different time intervals known as time slot at which data is transmitted at sender.



↳ Synchronous → A synchronous TDM is a technique where time slot is pre-assigned to every device

→ each device is given some time slot, irrespective of the fact that device contain the data or not.

→ Signal are send in the form of frames.

Subject: → 2. Asynchronous :- It is also known as / /
statistical TDM.

- Time slots are allocated to only those devices which have data to send,
- Each slot contain an address part that identify the source of the data.